

Section : Physics

Q.1 Which of the following is NOT correct in the case of the output obtained from a full wave rectifier?

1. It is unidirectional output.
2. It is pulsating output.
3. It is not intermittent output.
4. It is continuous output.

Options 1. 1

2. 2
3. 3
4. 4

Q.2 A long solenoid carrying a current produces a magnetic field B along its axis. If the value of current is halved and the number of turns per unit length is doubled then the new value of magnetic field is

1. $4B$
2. $2B$
3. $B/2$
4. B

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Match List-I with List-II

List-I

AC Circuit

(A) Phase Difference between current and voltage in a pure inductive ac circuit

(B) Phase Difference between current and voltage in a pure resistive ac circuit

(C) Phase Difference between current and voltage in a pure capacitive ac circuit

(D) Phase Difference between current and voltage in a series LCR circuit

List-II

Phase Angle

(I) zero

(II) $\pi/2$, current lags voltage

(III) $\tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

(IV) $\pi/2$, current leads voltage

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Two similar coils of radius r are placed concentrically with their planes at right angles to each other. The currents flowing in them are I and $3I$, respectively. The resultant magnetic induction at the centre is

- (1) $\frac{\sqrt{5}\mu_0 I}{2R}$
- (2) $\frac{\sqrt{10}\mu_0 I}{2R}$
- (3) $\frac{\sqrt{3}\mu_0 I}{2R}$
- (4) $\frac{\sqrt{3}\mu_0 I}{4R}$

Options 1. 1

2. 2
3. 3
4. 4

Q.5 An equiconvex lens has a focal length of 20 cm and its radius of curvature is 30 cm. Its refractive index is

1. $3/2$
2. $5/4$
3. $7/4$
4. $7/3$

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Two waves of amplitudes 3 m and 2 m, respectively superimpose at a point in the same phase. The resultant amplitude would be

1. 1.5 m
2. 5 m

3. 1 m
4. 6 m

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Monochromatic lights of wavelengths 620 nm and 626 nm, respectively are used to study diffraction at a single slit of aperture 0.7 mm. The distance between the slit and the screen is 1.8 m. The separation between the positions of the first maxima of the diffraction pattern obtained in the two cases is

1. 0.023 mm
2. 0.23 mm
3. 2.31 mm
4. 23.13 mm

Options 1. 1

2. 2
3. 3
4. 4

Q.8 A circular loop is placed in the plane of the paper in a magnetic field B . A current is induced in the loop in an anticlockwise direction. This is because the magnetic field B is

1. parallel to the plane of the loop and increasing with time
2. inward and decreasing with time
3. outward and decreasing with time
4. outward and increasing with time

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Two light beams of intensities in the ratio 25:9 are allowed to interfere. The ratio of the intensities of maxima and minima will be

1. 4:1
2. 5:3
3. 16:1
4. 625:81

Options 1. 1

2. 2
3. 3
4. 4

Q.10 In an AC generator, a 100 turns coil of area 0.10 m^2 rotates at 50 Hz in a uniform magnetic field of 0.01 T. The maximum voltage generated in the coil is

1. 27 V
2. 314 V
3. 31.4 V
4. 10 V

Options 1. 1

2. 2
3. 3
4. 4

Q.11 If 'n' identical resistors, each of value 'R' are first connected in series to a battery of emf 'E' and internal resistance 'R', the current drawn is 'I'. Now if they are connected in parallel, then the current drawn is '5 I'. The value of n is

1. 5
2. 6
3. 10
4. 25

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Identify the correct statements in the context of paramagnetic substances.

- (A) In external magnetic field B , individual atomic dipole moment points in opposite direction of B .
(B) Susceptibility is small and positive.
(C) In non-uniform magnetic field, the paramagnetic bar will tend to move from weak field to strong.
(D) Examples of paramagnetic substances are Aluminium, Sodium and oxygen.
Choose the correct answer from the options given below:

1. (B) and (C) only
2. (A), (B) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.13 An electric source has an emf of 130 V and a terminal voltage of 110 V. If it supplies a terminal current of 20 A, its internal resistance will be

1. 2 Ω
2. 1 Ω

3. 0.5Ω
4. 3Ω

Options 1. 1

2. 2
3. 3
4. 4

Q.14 An electric dipole is placed at an angle of 30° with an electric field of intensity $3 \times 10^5 \text{ N/C}$. It experiences a torque equal to 6 N m . If the dipole length is 2 cm , the value of charge will be

1. $5 \times 10^{-3} \text{ C}$
2. $8 \times 10^{-3} \text{ C}$
3. $6 \times 10^{-3} \text{ C}$
4. $2 \times 10^{-3} \text{ C}$

Options 1. 1

2. 2
3. 3
4. 4

Q.15 The different methods of producing plane polarised light are

- (A) Reflection
(B) Scattering
(C) Double Refraction
(D) Selective absorption

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Which of the following factors does Not affect the value of self inductance of a solenoid?

1. Number of turns
2. Area of cross-section of the solenoid
3. Permeability of the core material
4. current flowing through the inductor.

Options 1. 1

2. 2
3. 3
4. 4

Q.17 A 200 turns closely wound circular coil of radius 10 cm carries a current of 3.5 A . The magnetic moment of the coil is

1. 22 A m^2
2. 220 A m^2
3. 110 A m^2
4. 1.26 A m^2

Options 1. 1

2. 2
3. 3
4. 4

Q.18 Two point charges, $+2 \mu\text{C}$ and $-3 \mu\text{C}$, are separated by a distance of 20 cm in the air. At what point between the two charges and on the line joining the two charges is the electric potential zero? (Take the potential at infinity to be zero)

1. 10 cm from charge $+2 \mu\text{C}$
2. 6 cm from charge $+2 \mu\text{C}$
3. 12 cm from charge $+2 \mu\text{C}$
4. 8 cm from charge $+2 \mu\text{C}$

Options 1. 1

2. 2
3. 3
4. 4

Q.19 When ${}_{92}^{232}\text{Th}$ changes into ${}_{82}^{208}\text{Pb}$ the number of α and β particles emitted are

1. $\alpha = 6$; $\beta = 6$
2. $\alpha = 6$; $\beta = 4$
3. $\alpha = 4$; $\beta = 6$
4. $\alpha = 6$; $\beta = 2$

Options 1. 1

2. 2
3. 3
4. 4

Q.20 The frequency of a photon whose energy is 66.3 eV would be
(Given $h = 6.63 \times 10^{-34} \text{ Js}$)

1. $66.3 \times 10^{16} \text{ Hz}$
2. $3.32 \times 10^{16} \text{ Hz}$
3. $1.6 \times 10^{16} \text{ Hz}$
4. $8.9 \times 10^{16} \text{ Hz}$

Options 1. 1
2. 2
3. 3
4. 4

Q.21 Total internal reflection can take place only, if
1. the refractive indices of the two media are equal.
2. light goes from an optically denser medium to a rarer medium
3. light goes from an optically rarer medium to an optically denser medium
4. The angle of incidence is less than the critical angle.

Options 1. 1
2. 2
3. 3
4. 4

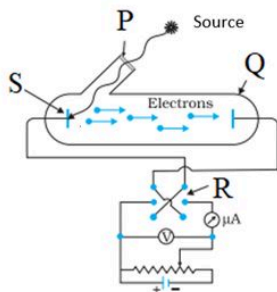
Q.22 In a parallel plate capacitor of capacitance C , a metal sheet is inserted between the plates parallel to them. If the thickness of the sheet is one third of the separation between the plates, the capacitance of the capacitor will be

1. $\frac{5C}{2}$
2. $2C$
3. $\frac{3C}{2}$
1. $5C$

Options 1. 1
2. 2
3. 3
4. 4

Q.23

In the experimental arrangement for study of photoelectric effect given below, P, Q, R and S, respectively are



- (A) Commutator
(B) Evacuated Glass Tube
(C) Photosensitive Plate
(D) Quartz Window

Choose the **correct** answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (B), (A), (C)
3. (B), (D), (A), (C)
4. (C), (B), (D), (A)

Options 1. 1
2. 2
3. 3
4. 4

Q.24 A galvanometer has a resistance of 8Ω . It is shunted by a wire of resistance 2Ω . If the total current is 2 A, the current passing through the shunt is

1. 1.6 A
2. 1 A
3. 2 A
4. 0.5 A

Options 1. 1

2. 2
3. 3
4. 4

Q.25 The following table gives the length, diameter and applied potential difference of four copper wires, A, B, C and D, across their ends. Arrange the wires in correct order of the increasing electric field.

difference	Length	Diameter	Potential
(A) Wire A	l	$3d$	V
(B) Wire B	$2l$	d	V
(C) Wire C	$3l$	$2d$	$2V$
(D) Wire D	$3l$	$2d$	V

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (D), (B), (C), (A)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.26 Two parallel plate capacitors, X and Y, have the same area of plates and the same separation between them. They are connected in series across a 12 V battery. X has air between the plates while Y contains a dielectric medium of dielectric constant 4. The ratio of electrostatic energy stored in X and Y is

1. 4:1
2. 1:4
3. 2:1
4. 1:2

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.27 Three charges Q, +2q and +2q are placed at the vertices of an equilateral triangle. If the net electrostatic potential energy of the system is zero, then Q is equal to

1. -q
2. +q
3. -2q
4. -q/2

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.28 A proton is projected with a speed of $4 \times 10^6 \text{ m s}^{-1}$ horizontally from east to west. A uniform magnetic field of strength $3 \times 10^{-3} \text{ T}$ exists in the vertically upward direction. The acceleration of the proton is (Given: mass of proton $1.67 \times 10^{-27} \text{ kg}$; charge on proton $1.6 \times 10^{-19} \text{ C}$)

1. zero
2. $11.5 \times 10^{11} \text{ ms}^{-2}$
3. $18.3 \times 10^{10} \text{ ms}^{-2}$
4. $32.1 \times 10^{11} \text{ ms}^{-2}$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.29 In a Young's double slit experiment, the fringe width is β . If the entire apparatus is immersed in a liquid of refractive index μ , the new fringe width is

1. β
2. $\beta \mu^{-2}$
3. $\beta \mu^{-1}$
4. $\beta \mu$

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.30 Arrange the following electromagnetic waves in the ascending order of their wavelengths.

- (A) Radio waves
- (B) Gamma rays
- (C) Infrared rays
- (D) Ultraviolet rays

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (D), (C), (A)
3. (B), (A), (D), (C)
4. (B), (D), (A), (C)

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Match List-I with List-II

List-I

Application

(A) Used in radiography

(B) Detection by stimulating nerve endings of the human retina

(C) Water purification

(D) Radar communication

List-II

Electromagnetic Waves

(I) UV rays

(II) Microwaves

(III) Visible light

(IV) X rays

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

4. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

Options 1. 1

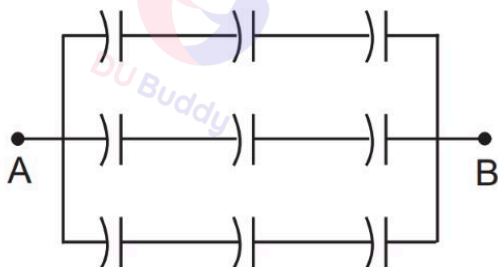
2. 2

3. 3

4. 4

Q.32

Each of the capacitors shown in the figure has a capacitance of $3\ \mu\text{F}$. The equivalent capacitance of the assembly between the points A and B is



1. $1\ \mu\text{F}$

2. $3\ \mu\text{F}$

3. $2\ \mu\text{F}$

4. $9\ \mu\text{F}$

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Two cells of emf 10 V and 6 V, each having internal resistances of $2\ \Omega$ are connected in parallel across points A and B. The equivalent emf of the cells is

1. 10 V

2. 8 V

3. 6 V

4. 4 V

Options 1. 1

2. 2

3. 3

4. 4

Q.34 A proton and an alpha particle of the same velocity enter in turn a region of uniform magnetic field, acting perpendicular to their direction of motion. The ratio of the radii of the circular paths described by the particles, respectively, would be

1. 1:2

2. 1:3

3. 2:1

4. 3:1

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Which of the following statements are correct related to electric potential energy?

(A) The electric potential energy is a vector quantity.

(B) The potential energy of two like charges is positive and for two unlike charges, it is negative.

(C) The work done by an electrostatic field in moving a charge from one point to another is independent of the path taken.

(D) The potential energy of an electric dipole in a uniform electric field is maximum, when the dipole moment is opposite to the electric field.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.36 The current through a resistor in a circuit is 0.6 A. On putting a resistance of $2\ \Omega$ in parallel with the resistor, the current drops to 0.9 A. The resistance of the resistor is

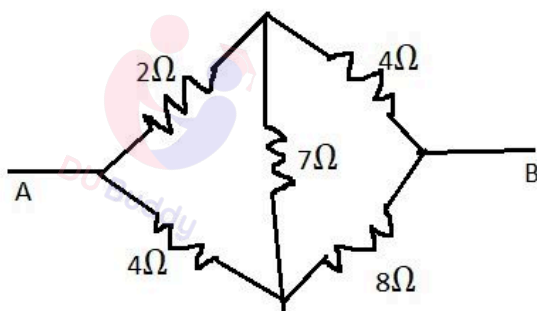
1. $1.0\ \Omega$
2. $2.0\ \Omega$
3. $4.0\ \Omega$
4. $8.0\ \Omega$

Options

1. 1
2. 2
3. 3
4. 4

Q.37

A bridge circuit is shown in the diagram. The equivalent resistance between A and B is



1. $4\ \Omega$
2. $2\ \Omega$
3. $8\ \Omega$
4. $6\ \Omega$

Options

1. 1
2. 2
3. 3
4. 4

Q.38 A nucleus with mass number 240 and binding energy per nucleon 7.6 MeV splits into two fragments each of mass number 120 with binding energy per nucleon 8.5 MeV. The amount of energy released in this process will be :

1. 0.9 MeV
2. 17 MeV
3. 216 MeV
4. 2040 MeV

Options

1. 1
2. 2
3. 3
4. 4

Q.39 Bulb B1 is a 50 W bulb working on 100 V and Bulb B2 is a 200 W bulb also working on 100 V. The ratio of resistances of bulbs B1 to B2 is:

1. 4:1
2. 4:5
3. 8:1
4. 2:1

Options

1. 1
2. 2
3. 3
4. 4

Q.40 The wavelength of photon emitted by a hydrogen atom when an electron makes a transition from $n = 2$ to $n = 1$ state is.

(Take the value of the Rydberg constant, $R = 1.0973 \times 10^7 m^{-1}$)

1. 121.5 nm

- 2. 194.8 nm
- 3. 490.7 nm
- 4. 913.3 nm

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.41 A plane electromagnetic wave of frequency 25 GHz is propagating in vacuum along the z-direction. At a particular point in space and time, the magnetic field is given by a value 5×10^{-8} T along Y-direction. The corresponding electric field is (speed of light $c = 3 \times 10^8$ m/s)

- 1. 15 V/m, along +X axis
- 2. 15 V/m, along -X axis
- 3. 1.66 V/m, along +X axis
- 4. 1.66 V/m, along -X axis

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.42 Choose a vector quantity from the following:

- 1. Electric charge
- 2. Electric potential
- 3. Electric dipole moment
- 4. Electric Flux

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.43 A resistor of resistance R, an inductor of $(20/\pi)$ H and a capacitor of $(5/\pi)$ μ F are connected in series to a voltage source of 220 V, 50 Hz. The value of current in the circuit is 4 A. The impedance of the circuit is

- 1. 40 Ω
- 2. 20 Ω
- 3. 45 Ω
- 4. 55 Ω

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.44 A transformer has 1000 turns in the primary and 2000 turns in its secondary winding. The primary voltage is 200 V and the load across the secondary is 100 Ω . If the transformer is assumed to be ideal, then the value of current in the primary is

- 1. 4.0 A
- 2. 2.5 A
- 3. 5.0 A
- 4. 8.0 A

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.45 Identify the correct statement among the following in the context of a nuclear reactor.

- 1. Coolant is used to slow down the neutrons.
- 2. Moderator is used to slow down the neutrons.
- 3. Cadmium rods are used to slow down the neutrons.
- 4. Energy released is of the order of keV.

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.46 From the following, choose the element for impurity atoms with which pure silicon should be doped to make it a p-type semiconductor:

- 1. Arsenic
- 2. Boron
- 3. Phosphorous
- 4. Antimony

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Q.47 The peak value of current in an ideal inductor of $(2/\pi)$ H inductance connected to a 200 V, 50 Hz ac supply will be

- 1. 1 A
- 2. 1.414 A
- 3. 1.5 A
- 4. 2 A

Options

- 1. 1

2. 2
3. 3
4. 4

Q.48 If a p-n junction is forward biased, then what is the direction of diffusion and drift current across the junction?

1. Diffusion - from n to p side, Drift-from p to n side
2. Diffusion - from p to n side, Drift -from n to p side
3. Diffusion - from n to p side, Drift -from n to p side
4. Diffusion - from p to n side, Drift-from p to n side

Options

1. 1
2. 2
3. 3
4. 4

Q.49 A tank is filled with water to a height of 15 cm. If the refractive index of water is 1.33, the apparent depth of a needle lying at the bottom of the tank will be

1. 11.28 cm
2. 12.6 cm
3. 10.8 cm
4. 13.24 cm

Options

1. 1
2. 2
3. 3
4. 4

Q.50 Identify the basic features of the photon picture of electromagnetic radiation from the following.

- (A) Each photon has energy $E = h\nu$ in vacuum.
- (B) Photons are electrically neutral and are not deflected by electric and magnetic fields.
- (C) All photons of light of a particular frequency ν or wavelength λ have the same energy and momentum.
- (D) If the intensity of light of a given wavelength is increased, there is an increase in the number of photons incident on a given area in a given time.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options

1. 1
2. 2
3. 3
4. 4